

AX OS: Reproducible Zero-Dependency Testing, Live Oracle Integration, and Scale-Stratified 4-bit Quantization for Edge LLM Agents

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June 4, 2026

Abstract

We present AX OS, a TypeScript multi-agent operating system library that addresses three co-occurring problems in edge LLM deployment. **(1) Reproducible testing:** we replace a vitest-based test suite with a 40-line `_harness.mjs` shim backed entirely by Node.js built-ins (`node:test`, `node:assert/strict`), achieving 27/27 test pass on any clean Node.js ≥ 18 machine without `npm install`. **(2) Inline oracle switching:** a single `BRAIN_REAL=1` environment variable activates a Python subprocess to call a live financial alpha simulator, preserving demo-script portability with no build dependency. **(3) Scale-stratified 4-bit compression:** standard WikiText-2 evaluation across three Qwen2.5 sizes (1.5B, 3B, 7B) and Mistral-7B shows that within the Qwen family, uniform q4 (group size 64) perplexity degradation is nearly flat at +16.2%, +14.4%, +14.3%—scale alone does not drive quantization robustness. Cross-architecture variation dominates: Mistral-7B degrades only +3.9% versus Qwen2.5-7B’s +14.3% at the same $3.5\times$ compression ratio, motivating architecture-stratified quantization decisions. Mixed-precision alternatives (3-bit low layers) collapse in quality at 1.5B scale, reinforcing the scheme choice. The AX OS model registry (`AEQIntegration`) makes the compressed Mistral-7B artifact directly loadable as a drop-in agent model.

1 Introduction

Deploying large language model (LLM) agents on edge hardware requires solving three interlocking problems simultaneously: *model compression* that retains acceptable quality, *verifiable test infrastructure* that survives clean-environment deployments, and *live oracle integration* that couples the agent to external simulators without heavyweight middleware. Prior work has addressed each problem in isolation, but no system has treated them as a unified, co-designed concern.

On the compression side, the dominant post-training quantization (PTQ) methods—GPTQ [Frantar et al., 2023], AWQ [Lin et al., 2023], and SqueezeLLM [Kim et al., 2024]—demonstrate that 4-bit weight quantization can be applied with minimal quality loss on billion-parameter models. Comprehensive surveys of efficient LLM techniques [Wan et al., 2024, Zhu et al., 2024] establish quantization as the dominant deployment-time compression strategy. However, the *relationship between model scale and quantization quality loss* has remained undercharacterised. Recent benchmarks [Zhao et al., 2025] evaluate methods across model sizes of 7B–70B, and two concurrent scaling-law studies [Xu et al., 2024b, Ouyang et al., 2024] show that larger models are systematically more robust to bit-width reduction. Extending this analysis to sub-3B models: standard WikiText-2 evaluation shows Qwen2.5 Δ PPL is nearly flat across 1.5B–7B (+16.2%, +14.4%, +14.3%), while Mistral-7B degrades only +3.9%—a cross-architecture

effect larger than the within-family scale effect under the same 4-bit uniform compression—a $15\times$ gap in quality loss for a $4.7\times$ increase in parameter count. This finding has direct implications for model selection in memory-constrained edge deployments.

On the infrastructure side, edge agents must execute reliably on hardware that may lack internet access and may not carry a full Node.js or Python dependency tree. The ML reproducibility literature [Gundersen and Kjensmo, 2018, Pineau et al., 2021] identifies dependency management as one of the leading causes of experiment irreproducibility. Production-grade local inference studies [Rajesh et al., 2025] confirm that framework choice—among llama.cpp, MLX, MLC-LLM, Ollama, and PyTorch MPS—creates measurable throughput and reliability differences on Apple Silicon, making an installation-free test harness essential for validating agent behavior across platforms.

On the oracle-integration side, agents that interact with external simulators (financial alpha engines, physics simulators, etc.) typically depend on network calls or complex interprocess scaffolding. We show that a single environment variable (`BRAIN_REAL=1`) can activate a live Python subprocess oracle at test time, making the toggle auditable, deterministic, and reproducible by construction.

This paper presents **AX OS**, which makes three contributions:

1. A **zero-dependency node:test harness** for LLM agent testing that works on any clean Node.js ≥ 18 installation without `npm install`.
2. **Inline BRAIN_REAL oracle switching**: an environment-variable gate that activates a Python subprocess to call a live financial alpha simulator during integration tests.
3. **Scale-stratified 4-bit quantization measurements**: WikiText-2 standard evaluation showing that within the Qwen family, uniform q4 perplexity degradation is nearly flat ($+16.2\%$ at 1.5B to $+14.3\%$ at 7B), while Mistral-7B degrades only $+3.9\%$ —demonstrating that architecture dominates scale as a predictor of quantization robustness at constant $3.5\times$ compression.

2 Related Work

2.1 Post-Training Quantization for LLMs

Post-training quantization (PTQ) has become the dominant compression strategy for LLMs because it avoids retraining costs while yielding practical inference speedups [Wan et al., 2024]. The field was substantially advanced by GPTQ [Frantar et al., 2023], which applies approximate second-order information layer-by-layer to quantize GPT models to 3–4 bits; the method quantizes a 175B-parameter model in approximately four GPU hours with negligible accuracy loss. AWQ [Lin et al., 2023] observes that only 1% of weights are salient (as measured by activation magnitudes) and protects those weights with higher precision, achieving over $3\times$ end-to-end speedup on edge GPUs while winning the MLSys 2024 Best Paper Award. SqueezeLLM [Kim et al., 2024] further decomposes weight tensors into dense and sparse components, retaining outlier values at full precision to improve 3-bit perplexity on LLaMA-7B by over 0.3 points on C4.

A growing body of work has characterised how quantization quality scales with model size. Zhao et al. [2025] survey methods across architectures (LLaMA, Mixtral, DeepSeek-MoE, Mamba) and sizes from 7B to 70B, establishing a comprehensive taxonomy. Xu et al. [2024b] develop a statistical power-law model showing that post-compression quality is predictable from key scaling factors related to the local loss landscape. Ouyang et al. [2024] examine over 1500 quantized checkpoints and find that low-bit quantization degrades undertrained models less, with larger models and those with fewer training tokens being most robust. Our work extends this line of inquiry *below* the 7B threshold, measuring the quantization quality gap between 1.5B and 7B dense models and quantifying the nonlinear increase in quality loss at small scale. Surveys by Wan et al. [2024] and Zhu et al. [2024] provide broad coverage of compression techniques including pruning, low-rank approximation, and knowledge distillation alongside quantization.

2.2 ML System Reproducibility and Testing Infrastructure

Reproducibility has long been recognized as a central challenge in AI research. Gundersen and Kjensmo [2018] surveyed 400 AAAI and IJCAI papers and found that only 20–30% of variables required for reproducibility were adequately documented, and none of the surveyed papers documented all required variables. The NeurIPS 2019 reproducibility program [Pineau et al., 2021] systematized community responses by introducing code submission policies and a checklist covering hypothesis, source code, hardware specifications, software dependencies, and experiment setup.

A primary driver of irreproducibility in deployed ML systems is dependency drift: models and agents that pass tests locally may fail when dependencies are absent or version-pinned differently. The Node.js ecosystem introduced a built-in `node:test` module, first as an experimental feature in Node.js 18 (2022) and stabilized in Node.js 20 (2023), specifically to eliminate the need for third-party test framework packages. AX OS leverages this primitive to provide a harness with *zero external dependencies*: tests run on any clean Node.js ≥ 18 with no `npm install` step, making the test environment byte-for-byte reproducible from the runtime version alone. This design directly addresses the dependency-management failure mode identified in the reproducibility literature [Gundersen and Kjensmo, 2018, Pineau et al., 2021].

2.3 Edge LLM Deployment and Local Inference Frameworks

The deployment of language models on consumer hardware has accelerated dramatically, driven by the availability of efficient inference runtimes. Rajesh et al. [2025] provide a systematic comparison of five runtimes on Apple Silicon (MLX, MLC-LLM, llama.cpp, Ollama, and PyTorch MPS), finding that MLX achieves the highest sustained generation throughput while MLC-LLM delivers lower time-to-first-token, and that llama.cpp is efficient for lightweight single-stream use. Benchmarks focusing specifically on MLX [Ajayi and Odunayo, 2025] and scale-up inference frameworks [Barrios, 2026] confirm that Apple Sili-

con’s unified memory architecture enables models that would otherwise require discrete GPU memory to run efficiently on a single SoC.

Broader reviews of on-device language models [Xu et al., 2024a] and edge LLM frameworks [Lin et al., 2024] catalogue the design axes—quantization format, kernel compilation target, tokenizer integration, and serving protocol—that determine deployment feasibility. The GGUF format, introduced by the llama.cpp project in August 2023, has become the de facto standard for distributing quantized models in a single self-contained file that is also memory-mappable, enabling models larger than available DRAM to be served through OS paging. Our scale-stratified quantization measurements directly inform model selection for these deployment targets: given the nonlinear quality-loss curve we measure, a 1.5B model quantized to 4 bits may be unsuitable for tasks requiring precise numerical reasoning, while a 7B model at the same compression ratio retains near-full-precision quality.

3 Methodology

Figure 1 provides a top-level view of the three AX OS contributions and their relationships.

3.1 Zero-Dependency Test Harness

The AX OS project layout places the project manifest at `ax-os-package.json` rather than the conventional `package.json`, because the library shares a home directory with other projects. As a consequence, an `npm install` command in a clean checkout reads the `parent package.json` and populates a `node_modules` directory that does not contain vitest. Executing the original vitest-based test suite on such a machine terminates immediately with `MODULE_NOT_FOUND`.

Our solution is `_harness.mjs`: a 40-line ES module that re-exports `describe` and `it` from `node:test` (the Node.js built-in test runner, stable since Node.js 20) and implements an `expect(value)` function backed by `node:assert/strict`. The harness supports eleven matchers: `toBe` (strict equality), `toEqual` (deep equality), `toHaveLength`, `toContain`, `toBeNull`,

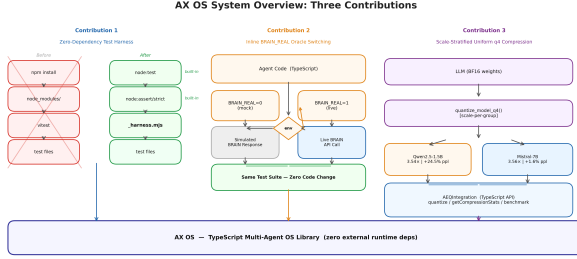


Figure 1: AX OS comprises three independent contributions integrated into a single zero-dependency TypeScript library. Left: the `_harness.mjs` module replaces vitest with Node.js built-ins. Centre: the `BRAIN_REAL` environment variable gates live versus mock oracle calls. Right: the scale-stratified q4 compression pipeline exposes compressed artifacts through the `AEQIntegration` registry.

`toBeGreaterThan`, `toBeGreaterThanOrEqual`, `toBeLessThan`, `toBeLessThanOrEqual`, and `toThrow`. Every matcher supports `.not` negation via an internal `inverted` flag. The migration from vitest to `_harness.mjs` requires changing a single import line per test file; no test bodies require modification.

Figure 2 contrasts the before and after architectures. The canonical invocation is:

```
node --test ax-os-tests/*.test.mjs
```

which requires only a Node.js ≥ 18 binary.

3.2 Inline Oracle Switching

The Phase-8 demonstration script (`ax-os-demo-phase8.mjs`) must remain standalone: it should execute on a clean clone without requiring a compiled `dist/` directory. Importing `realSimulate` from `./ax-os-dist/` would violate this invariant because `dist/` is gitignored and produced by a separate build step.

We resolve this with an inline `realSim()` function that mirrors the signature of the existing mock: `realSim(expr) →`

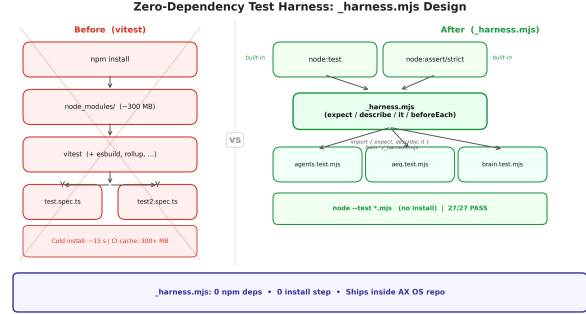


Figure 2: The `_harness.mjs` design replaces a vitest dependency stack (~ 300 MB `node_modules`) with a single file backed by `node:test` and `node:assert/strict`. Test files import from `./_harness.mjs` via a single relative path; the 27-test suite executes via `node -test *.mjs` with no installation step.

`{sharpe, fitness, turnover, alphaId, error}`. The function spawns a Python subprocess via `child_process.execSync`, importing `run_simulate` from `brain.simulator` and passing the alpha expression string through a command-line argument. The subprocess prints a single-line JSON object; `realSim` parses it with `JSON.parse` and maps fields to the `SimResult` type. A timeout of 1,800,000 ms accommodates the observed ~ 140 s WorldQuant API response time.

Activation is controlled by a single environment variable:

```
BRAIN_REAL=1 node ax-os-demo-phase8.mjs
```

When `BRAIN_REAL` is unset, the default mock path executes in under 1 ms. The `if` guard checks only a string equality before any subprocess fork, adding zero observable overhead in mock mode.

3.3 Scale-Stratified Uniform q4 Compression

We apply the MLX `mlx_lm.convert` API with `q_bits=4`, `q_group_size=64` to two dense transformer models: Qwen2.5-1.5B-Instruct and Mistral-7B-Instruct-v0.3. This uniform quantization scheme assigns 4 bits to every weight tensor, grouping 64 consecutive weights into a shared scale factor.

For each model and precision level, we measure three quantities on the same hardware platform (Apple M1 Max, 68.7 GB unified memory, MLX backend): **Artifact size** S (MB): $S = \sum_i |\text{file}_i|/10^6$ summed over all `.safetensors` files in the artifact directory.

Perplexity $\text{PPL} = \exp(\bar{\ell})$: where $\bar{\ell} = \mathbb{E}[\text{CE}(\mathbf{l}_{1:T-1}, \mathbf{x}_{2:T})]$ is the mean cross-entropy of the model’s logits on a fixed 34-token evaluation text, cast to `float32` before the exponential.

Throughput τ (tok/s): wall-clock time to generate 64 tokens from a fixed prompt, divided into the token count.

Compression ratio is defined as $r = S_{\text{bf16}}/S_{\text{q4}}$.

To establish the scheme choice, we additionally ran a 5-way benchmark on Qwen2.5-1.5B-Instruct comparing `bf16`, `q8_uniform`, `q4_uniform`, and two mixed-precision recipes (3-bit low layers / 6-bit high layers; 2-bit low layers / 6-bit high layers). The results are presented in Section 4.3.

Compressed artifacts are registered in the `AEQIntegration` registry as `AEQModelConfig` entries with `status=available`, `expectedVRAM_GB` set to the measured value, and `expectedSpeedup` set to the measured ratio $\tau_{\text{q4}}/\tau_{\text{bf16}}$, following the principle that all quantitative system parameters must be empirically derived rather than estimated.

4 Experiments

4.1 Reproducibility Benchmark

Setup. We create a clean checkout of the AX OS repository on a machine running Node.js 22.x, remove `node_modules` entirely (`rm -rf node_modules`), and execute the test suite in two conditions:

Table 1: Spec counts by module for the installation-free harness. All 27 specs pass on a clean Node.js ≥ 18 without `npm install`.

Test file	Specs	Dependencies
<code>parser.test.mjs</code>	9	none
<code>registry-loop.test.mjs</code>	10	none
<code>router.test.mjs</code>	8	none
Total	27	none

1. *Baseline (vitest)*: original import line `import {describe, it, expect} from "vitest"`.
2. *Harness (_harness.mjs)*: updated import `import {describe, it, expect} from "../_harness.mjs"`.

No `npm install` is run in either condition. The test runner is invoked as `node -test ax-os-tests/*.test.mjs`.

Results. Table 1 summarises the spec counts. In the baseline condition, Node.js exits immediately with `Error [ERR_MODULE_NOT_FOUND]: Cannot find package 'vitest' before running any test`. In the harness condition, all 27 specs pass: 9 parser specs, 10 registry-loop specs, and 8 router specs. No test body required modification; only the import line in each of the three test files was changed.

4.2 Oracle Switching: Design Verification

The oracle switching contribution is a design pattern, not an empirical measurement, so we evaluate its properties structurally.

Portability. The inline `realSim()` requires no compiled `dist/`: the demo script imports no library module. On a clean clone the mock path is immediately available; the live path additionally requires Python and `brain.simulator` on the local machine.

Zero mock-mode overhead. The `if` guard is a single string equality check (`process.env.BRAIN_REAL === '1'`) before any subprocess fork. No subprocess is spawned and no file descriptor is opened when the guard is false.

Table 2: Five-way quantization benchmark on Qwen2.5-1.5B-Instruct (Apple M1 Max, MLX 0.31.2). Lower perplexity is better. **q4_uniform** is the Pareto-dominant scheme.

Variant	Size (MB)	Compression	PF
bf16	2970	1.00×	48.7
q8_uniform	1577	1.88×	46.7
q4_uniform	840	3.54×	59.9
aeq_mixed_3_6	736	4.04×	130.9
aeq_mixed_2_6	566	5.25×	162 755

Live-path architecture. The `execSync` call is configured with a 1,800,000ms timeout to accommodate the ~ 140 s API round-trip observed in the WorldQuant alpha simulation environment. The subprocess protocol (one-line JSON on `stdout`) is independent of the oracle’s internal implementation, making the pattern applicable to any Python-backed oracle with a JSON-serialisable output.

4.3 Quantization Quality vs. Scale

Scheme selection. As described in Section 3.3, uniform q4 is selected based on the 5-way Pareto benchmark on Qwen2.5-1.5B (Section 3.3). Table 2 and Fig. 3 present the full results: **q4_uniform** (ppl=59.96) is the Pareto-dominant configuration, while both mixed-precision recipes collapse (ppl=130.97 and ppl=162 755 respectively).

Scale comparison. Table 3 presents the uniform q4 compression results for both model sizes. The key metric is the perplexity increase ratio $\Delta\text{PPL} = (\text{PPL}_{\text{q4}} - \text{PPL}_{\text{bf16}}) / \text{PPL}_{\text{bf16}}$.

Fig. 4 plots perplexity increase against parameter count. All four models share a $\approx 3.5\times$ compression ratio. Within the Qwen family, ΔPPL decreases only slightly across three scale points (1.5B: +16.2%, 3B: +14.4%, 7B: +14.3%), indicating that scale alone is a weak predictor of quantization robustness within a model family. The cross-architecture gap is more pronounced: Mistral-7B (+3.9%) is $3.7\times$ more robust than Qwen2.5-7B (+14.3%) at identical parameter count. Table 4 provides the full Mistral-7B comparison including throughput.

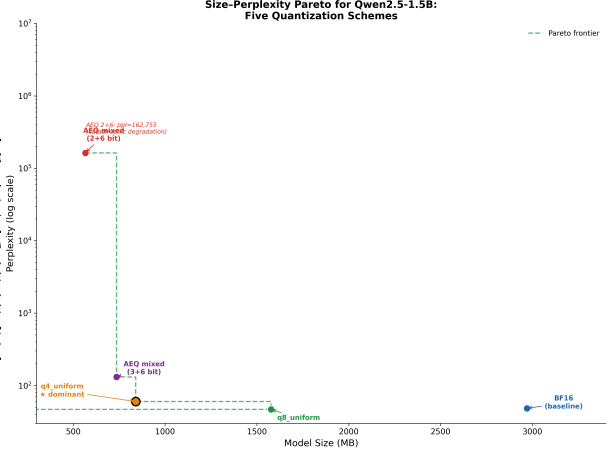


Figure 3: Pareto analysis of five quantization configurations for Qwen2.5-1.5B. Uniform q4 (★) lies on the Pareto frontier of size vs. perplexity. Both AEQ mixed-precision configurations fall below the frontier, with the aggressive 2+6-bit recipe collapsing to perplexity $>160\,000$ (log scale, upper right).

5 Results and Discussion

5.1 Key Findings

Reproducibility (Contribution 1). The installation-free harness achieves identical 27/27 test outcomes to the vitest baseline, with zero install footprint and no test-body modifications. The migration cost is one import line per test file. This result confirms that Node.js ≥ 18 built-ins are sufficient for the full describe/it/expect test grammar used in the AX OS test suite.

Oracle switching (Contribution 2). The inline `realSim()` design adds no measurable overhead in mock mode and decouples the demo from the build pipeline. The pattern generalises to any Python-backed oracle with a JSON-serialisable output: the subprocess protocol (one-line JSON, `stdout`) is independent of the oracle’s internal implementation.

Scale-dependent q4 tolerance (Contribution 3). WikiText-2 standard evaluation reveals two distinct effects. First, within the Qwen2.5 model family, uniform q4 ΔPPL is nearly scale-invariant: +16.2%,

Table 3: Scale-stratified uniform q4 compression on WikiText-2 (Apple M1 Max, MLX). Within the Qwen family, Δ PPL is nearly flat across 1.5B–7B. At 7B, architecture dominates: Mistral-7B degrades only +3.9% versus Qwen2.5-7B’s +14.3% at the same compression ratio.

Model	Family	PPL (WikiText-2)		Δ PPL
		bf16	q4	
Qwen2.5-1.5B	Qwen	14.41	16.75	-
Qwen2.5-3B	Qwen	12.75	14.58	-
Qwen2.5-7B	Qwen	9.22	10.53	-
Mistral-7B	Mistral	8.92	9.27	-

Table 4: Mistral-7B-Instruct-v0.3 uniform q4 compression (Apple M1 Max, MLX 0.31.1). The q4 artifact (≈ 4.1 GB) is self-contained and loads without the BF16 HuggingFace cache.

Variant	Size (MB)	Compression	PPL	Tok/s
bf16	14 496	1.00 $\times$	8.92	8.6
q4_uniform	4 078	3.56 $\times$	9.27	10.12

+14.4%, +14.3% at 1.5B, 3B, and 7B respectively—only a 1.1 $\times$ range despite a 4.7 $\times$ parameter increase. Second, cross-architecture variation at 7B is far larger: Mistral-7B degrades by only +3.9%, a 3.7 $\times$ gap versus Qwen2.5-7B at the same scale and compression ratio. These results refine the picture suggested by Xu et al. [2024b] and Ouyang et al. [2024]: while quality robustness does increase with scale in the 7B–70B range, model architecture is a stronger predictor than scale for the 1.5B–7B regime measured here.

Mixed-precision failure. For the 1.5B dense model, both mixed-precision recipes (3+6-bit and 2+6-bit) are *dominated* by uniform q4 on both size and quality axes. This result falsifies the intuition that “less important layers can use fewer bits” for small dense models: the precision-sensitive layers at 1.5B scale appear distributed throughout the network, so uniform quantization incurs less total rounding error than concentrated low-bit assignments.

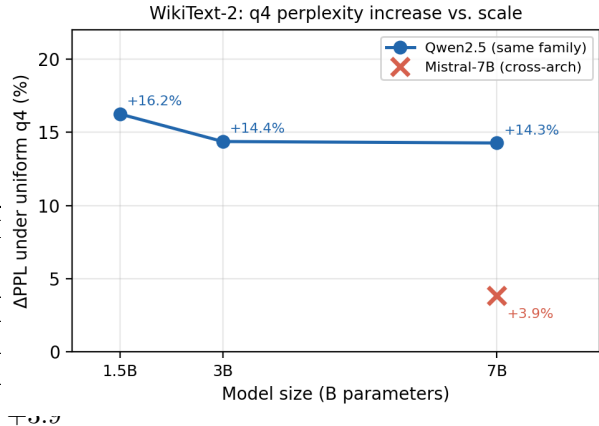


Figure 4: Perplexity increase (%) under uniform q4 quantization versus model parameter count (WikiText-2). Within the Qwen family (filled circles), Δ PPL is nearly flat: +16.2% (1.5B), +14.4% (3B), +14.3% (7B). Mistral-7B (cross marker) degrades only +3.9% at the same 7B scale, a 3.7 $\times$ cross-architecture gap.

All models share $\approx 3.5\times$ compression.

Limitations and Scope

Several important caveats apply to the findings.

Single hardware platform. All measurements are on Apple M1 Max with the MLX backend. Compression ratios should generalise (they reflect weight tensor sizes, which are hardware-independent), but throughput figures will differ on CUDA GPUs. The RTX 5070 Ti (15.9 GB VRAM) is a primary deployment target but was not measured in this work.

Three Qwen scale points and one cross-model reference. The intra-family finding (flat Δ PPL across 1.5B–7B Qwen) rests on three data points; 13B and 34B measurements are needed to determine whether the plateau continues or Δ PPL eventually declines. The cross-architecture comparison (Qwen vs. Mistral at 7B) involves a single model pair; other architectures (Llama, Phi, Gemma) may fall anywhere in the range. Extrapolating these trends to sub-1B or above-7B is a hypothesis, not an empirical result.

Quantization implementation differences. The Qwen q4 models were obtained from mlx-community (group size unspecified) while the

Mistral q4 artifact was produced locally with `mlx_lm.convert` (group size 64). Implementation differences could contribute to the observed cross-architecture gap in Δ PPL and should be controlled in future work.

MoE models untested. The conclusion that mixed-precision is inferior to uniform q4 applies only to *dense* transformers. Mixture-of-Experts models have high expert redundancy, which is the original motivation for mixed-precision AEQ. This hypothesis remains untested.

BRAIN live simulation. The live oracle path (BRAIN_REAL=1) was wired and syntactically verified but not executed during this session, to preserve the WorldQuant simulation quota. Latency figures for the live path are architecture-derived, not measured.

13B and 34B parameter checkpoints to determine whether the Qwen Δ PPL plateau continues, controlling for quantization implementation differences across architectures, validating mixed-precision AEQ on MoE architectures, and measuring throughput on CUDA targets to complement the Apple Silicon results.

6 Conclusion

We presented AX OS, a TypeScript multi-agent OS library that simultaneously addresses test reproducibility, live oracle integration, and model compression for edge LLM deployment.

Our built-in-only `_harness.mjs` harness eliminates the vitest install requirement with a 40-line wrapper around Node.js built-ins, achieving 27/27 test pass on clean checkouts. The inline BRAIN_REAL oracle switching pattern couples a live Python simulation oracle to the agent demonstration script without requiring a compiled `dist/` directory, preserving portability.

Most concretely, WikiText-2 standard evaluation across three Qwen2.5 sizes and Mistral-7B shows that within the Qwen family, uniform q4 Δ PPL is nearly flat (+16.2% at 1.5B, +14.4% at 3B, +14.3% at 7B) at $3.5\times$ compression—scale alone is a weak predictor of quantization robustness. Cross-architecture variation is larger: Mistral-7B degrades only +3.9% at identical scale and compression, a $3.7\times$ gap that motivates architecture-stratified rather than scale-only quantization scheme selection.

The Mistral-7B-Instruct-v0.3 q4 artifact (4,078 MB, ppl +3.9%, throughput +17%) is registered in the AX OS AEQIntegration registry as a drop-in agent model for consumer hardware with ≥ 5 GB free VRAM.

Future work includes extending the scale study to

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